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Guidance

Safety Bulletin 23 Aerosol Fire Suppression Systems – Incorrect Wiring Arrangements

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Contents

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1. Summary of Issue

Following a recent incident on board a workboat where a fixed aerosol fire suppression system activated unintentionally and without warning or human input, it has become evident that the wiring arrangement of the system was not as prescribed in the manual and approved components were not used. This safety bulletin reinforces the importance of installing the system as dictated by the manufacturer's instructions.

Fixed condensed aerosol fire suppression systems (such as FirePro, Stat-X and MAG/Pyrogen) operate by a pyrotechnic charge initiating a controlled burn in a solid compound which releases the aerosol medium into the protected space. The pyrotechnic charge is ignited by a current supplied from the control system.

In the installation in question, the control cabling for the fixed aerosol fire suppression system was bundled on the same cable tray with many other cables. There was a mix of alternating and direct current cables in the same cable tray alongside the control cable for the fixed condensed aerosol fire suppression system. This mixture of cables (including power cables) is prohibited by the manufacturer as it may cause an induced current in the control cable.

Due to the low current required to ignite the pyrotechnic charge on the system, it is possible that an induced current may have been responsible for a recent erroneous activation of the fixed condensed aerosol fire suppression system.

2. Actions to Take

Owners of vessels with fixed condensed aerosol fire suppression systems installed should ensure that the control cabling and all other cabling associated with the system is installed in line with the manufacturer's instructions as well as the applicable standards. The control cable should not be in the same cable tray, conduit or other cable routing as power cables or any other alternating current cables unless it is properly shielded.

Vessel owners should check the installation onboard and not rely on the installation drawings to ensure the cabling is as required.

If the control cable is found to be run in the same cable tray, conduit or other cable routing as power or other AC cables and is not correctly shielded a remedial action plan should be developed to reroute the control cable and this work should be done at the earliest practicable opportunity. This should be undertaken by a suitably qualified person and should be checked by a service engineer approved by the manufacturer.

Further to the above the following general requirements for fixed aerosol fire suppression systems are reiterated to help improve safety.

The condensed aerosol fixed fire suppression systems are to be:

- Installation & Commissioning

Installed and commissioned by an authorised competent person, using approved components, in accordance with the manufacturer's instructions / manual/s and conditions of approval.

This includes manufacturer's instructions regarding the use of the correct cabling, avoiding routes which could induce current, shielding and system isolation arrangements to ensure that the system is only activated when intended.

- Annual Service

Serviced annually by the authorised competent person in accordance with the manufacturer's instructions / manual/s and conditions of approval

- Monthly Inspection

Inspected monthly by the operator in accordance with the manufacturer's instructions / manuals and conditions of approval.

- Safe Entry to the protected Space

Made safe for personnel to enter the protected space. Prior to entry isolation switches, where provided, shall be activated in accordance with the manufacturer's instructions / manuals and conditions of approval.

Risk assessments shall be completed in relation to the hazards envisaged with the system.

- Documentation

The vessel is to retain documentation onboard relating to the operation and maintenance of the system.

Commissioning checklists need to be completed and valid for the actual system as installed.

Records of installation, commissioning, servicing, and inspections are to be retained onboard the vessel. It is recommended that aide memoires / checklists be developed and circulated by the manufacturer so that the system is installed, commissioned, and maintained to the manufacturers requirements and conditions of approval.

3. Further Information

Any queries should be directed to the MCA at

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